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Problem #01

A heat engine receives 6kW from a 250°C source and rejects heat at 30°C. Examine each of the following three cases with respect to the inequality of Clausis

(a) $W = 6\text{kW}$

$W = 6000 \text{ W}$

$Q_H = W + Q_L$

$$\oint Q/T = \frac{Q_H}{T_H} - \frac{Q_L}{T_L}$$

$6 = 6 + Q_L$

$Q_L = 0$

$$\oint Q/T = \frac{6{,}000}{250+273} - \frac{0}{30+273}$$

$$\oint Q/T = 11.4722 \text{ W/K}$$

According to Clausis inequality $\oint Q/T > 0$, the cycle is impossible.

(b) $W_o = 0\,kW$

$Q_H = W + Q_L$

$6 = 0 + Q_L$

$Q_L = 6{,}000\,W$

$$\oint \frac{Q}{T} = \frac{Q_H}{T_H} - \frac{Q_L}{T_L}$$

$$= \frac{6{,}000}{523} - \frac{6000}{303}$$

$$\oint \frac{Q}{T} = -8.3297\ W/K$$

$\oint \frac{Q}{T} < 0$, The cycle is possible and irreversible.

(iii) Carnot Cycle

For carnot cycle efficiency is given by $\eta = 1 - \frac{Q_L}{Q_H} = 1 - \frac{T_L}{T_H}$

$$\frac{Q_L}{Q_H} = \frac{T_L}{T_H}$$

$$\frac{Q_L}{T_L} = \frac{Q_H}{T_H}$$

$$\frac{Q_L}{303} = \frac{6{,}000}{523}$$

$Q_L = 3476.1\ W$

$Q_L = 3.4761\ kW$

The work output is

$Q_H = W + Q_L$

$6 = W + 3.4761$

$W = 2.5239\ kW$

$$\oint \frac{Q}{T} = \frac{Q_H}{T_H} - \frac{Q_L}{T_L}$$

$$\oint \frac{Q}{T} = 0\ W/K$$

The cycle is possible and reversible

Problem #02

In a carnot engine with ammonia as the working fluid, the high temperature is 60°C and as Q_H is received, the ammonia changes from saturated liquid to saturated vapor. The ammonia pressure at low temperature is 190 kPa. Find T_L, the cycle thermal efficiency, heat added per kilogram, and the entropy s, at the beginning of the heat rejection process.

(i) $T_L = -20°C$

(ii) Cycle Thermal efficiency $= 0.24$

(iii) Heat added per kg $= 996.5$ kJ/kg

(iv) Entropy at beginning of heat rejection $= 4.6577$ KJ/k

(i) $T_H = 60°C = 333\ K$

$P_L = 190\ kPa$

Using saturated ammonia table

$P = 190\ kPa$

$T = -20°C$

$T_L = 253\ K$

(ii) Cycle Thermal efficiency

$$\eta = 1 - \frac{T_L}{T_H}$$

$$\eta = 1 - \frac{253}{333}$$

$$\eta = 0.24$$

(iii) Heat added per kg = ?

From saturated ammonia table

$T = 60^\circ C$

Specific entropy for saturated Liquid

$S_f = S_1 = 1.6652$ KJ/kg·K

Specific entropy for saturated liq vapor

$S_g = S_2 = 4.6577$ KJ/kg-k

∴ Heat added per unit kg.

$q_H = \int T \cdot ds = T(S_2 - S_1)$

$= 333(4.6577 - 1.6652)$

$q_H = 996.5$ KJ/kg

(iv) Entropy at the beginning of heat rejection process

From saturated ammonia table for $T = 60^\circ C$

$S_3 = S_2 = S_g(60^\circ C) = 4.6577$ KJ/Kg-K

Problem #03

$T_{H,H} = 200°C = 473\ K$

$P_{L,H} = 20\ KPa$

$T_{H,R} = 20°C = 293\ K$

$T_{L,R} = -15°C = 258\ K$

At $T_{L,H} = 200°C$ From saturated water Table

$h_f = 852.26\ KJ/kg$

$h_g = 2792\ KJ/kg$

$q = h_g - h_f$

$q = 2792 - 852.26$

$q = 1939.74\ KJ/kg$

$T_{L,H} = T_{sat}$ at $20\ KPa = 60.06°C = 333.06\ K.$

$$\eta = 1 - \frac{T_{L,H}}{T_{H,H}}$$

$$\eta = 1 - \frac{333.06}{473}$$

$$\eta = 0.296$$

$$COP_R = \frac{T_{LR}}{(T_{H,R} - T_{L,R})} , = \frac{258}{293-258} , = 7.37$$

$$COP_R = \frac{Q_L}{W_{in,R}}$$

$$W_{in,R} = \frac{Q_L}{COP_R}$$

$$W_{in,R} = \frac{1}{7.37}$$

$$= 0.136 \text{ kJ}$$

$$W_{out,H} = W_{in,R} = 0.136 \text{ kJ}.$$

$$\eta = \frac{W_{out,H}}{Q_{H,H}}$$

$$Q_{H,H} = \frac{W_{out,R}}{\eta}$$

$$Q_{H,H} = \frac{0.136}{0.296}$$

$$\boxed{Q_{H,H} = 0.46 \text{ kJ}}$$